Irtaza Ahmed

Machine Learning Developer

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SUMMARY

Highly motivated and results-driven Machine Learning Developer and Academic Gold Medalist with a proven track record of engineering high-accuracy predictive pipelines and advanced ensemble architectures. Proficient in cross-validation paradigms, complex data imputation (MICE), and custom stacking frameworks spanning diverse domains from tabular analytics to molecular drug discovery research. Strong foundation in computer science principles, combining rigorous statistical engineering with scalable analytical workflows.

SKILLS

- **Machine Learning:** Supervised Learning, Classification, Regression, Ensemble Methods, Soft/Hard Voting, Stacking, Nested Classifiers, XGBoost, LightGBM, CatBoost, Random Forest etc.
- **Data Engineering:** Data Preprocessing, MICE Imputation, Regex Pipelines, High-Cardinality Encoding, Feature Engineering, Cross-Validation (k-Fold)
- **Languages & Tools:** Python, C++, Git, GitHub, Jupyter Notebook, Google Colab, Figma
- **Frameworks/Libs:** Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn

EDUCATION

Bachelor of Science in Computer Science (Lateral Entry in BS-CS) | Bahria University Lahore Campus

- **Expected Graduation:** 2027
- **Academic Standing:** Current CGPA: **3.60**/4.00
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Operating Systems, Compiler Construction, Theory of Automata, Human-Computer Interaction, Probability & Statistics.

Associate Degree Program in Computer Science (ADP-CS) | Bahria University Lahore Campus

- **Graduated:** June 2025
- **Academic Achievements:** Awarded **Gold Medal** for Academic Excellence
- **Graduation CGPA: 3.84**/4.00 (Batch Rank: **1st**)

PROJECTS

Molecular Activity Prediction for Drug Discovery | ML Research Project (In Progress)

- **Pipeline Engineering:** Developing a robust, high-performance predictive pipeline to classify molecular activity profiles (active vs. inactive compounds) targeting automated drug discovery workflows.
- **Model Benchmarking:** Implemented and benchmarked foundational baseline architectures including Logistic Regression, Support Vector Machines (SVM), and Decision Trees utilizing both standard partitions and rigorous 5-Fold Cross-Validation loops.
- **Hierarchical Ensembling:** Engineering an advanced, custom hierarchical Nested Stacking Classifier and Voting frameworks fusing traditional statistical estimators with state-of-the-art Gradient Boosting algorithms (XGBoost, LightGBM, CatBoost).
- **Evaluation Optimization:** Optimizing complex model evaluation loops to enforce maximum structural generalization, fine-tuning classification thresholds to balance precision-recall metrics for low-frequency active molecule identification.
- **Tech Stack:** Python, Scikit-Learn, XGBoost, LightGBM, CatBoost, Ensemble Learning, Pandas, NumPy.

Chronic Kidney Disease Prediction System | Ensemble Learning Classification

- **Data Imputation & Imbalance:** Applied Multivariate Imputation by Chained Equations (MICE) to resolve missing features in medical records and tuned structural parameters to counter severe dataset class imbalances.
- **Ensemble Architecture:** Engineered a high-accuracy Soft-Voting Classifier fusing XGBoost, Random Forest, and Logistic Regression to maximize target recall for critical clinical classifications.
- **Explainability:** Extracted and visualized global feature importances using Scikit-Learn to isolate key diagnostic metrics (such as GFR and Serum Creatinine) for objective clinical validation.
- **Tech Stack:** Python, Google Colab, Scikit-Learn, XGBoost, Pandas, NumPy, Seaborn, Matplotlib.

Used Car Price Valuation Engine | Ensemble Regression Pipeline

- **RegEx Processing Pipeline:** Developed a custom string cleaning workflow using regular expressions (RegEx) to transform raw, noisy textual currencies and mileages into structural float arrays.
- **High-Cardinality Encoding:** Implemented dynamic frequency-mapping for extreme categorical dimensions (such as vehicle brand and model) alongside dummy variables to prevent dimensionality explosion.
- **Regression Modeling & Benchmarking:** Constructed a robust unified Voting Regressor combining XGBoost and Random Forest architectures to map continuous vehicle asset depreciation trends.
- **Performance:** Achieved a top-tier R^2 Score of 0.9429, accurately accounting for 94.2% of underlying target pricing variance on unseen out-of-distribution test partitions.
- **Tech Stack:** Python, Google Colab, Scikit-Learn, XGBoost, Pandas, NumPy, Seaborn, Matplotlib.

Telecom Customer Churn Prediction Engine | Ensemble Classification Network

- **Automated Data Cleansing:** Automated an end-to-end cloud pipeline to ingest web-streamed user profiles, repairing implicit string alignment defects, noisy features, and complex type coercions.
- **Stratified Tree Optimization:** Leveraged stratified train-test splits and specialized loss weights (scale_pos_weight) within tree frameworks to handle heavily skewed user retention data distributions.
- **Predictive Pipeline:** Blended an optimized XGBoost Classifier and Random Forest into a production-grade soft-voting network to construct robust, non-linear classification boundaries.
- **Business Evaluation:** Achieved a global ROC-AUC Score of 0.8443 and successfully flagged 80% of historical customer churn instances (Recall) to actively support automated enterprise retention workflows.
- **Tech Stack:** Python, Google Colab, Scikit-Learn, XGBoost, Pandas, NumPy, Seaborn, Matplotlib.